Choc Cycle!

Ingredients:

Crunchie bar

Marble effect chocolate

Lion bar

Large bars of; milk chocolate and white chocolate

Cheese grater

Large clear mixing bowl

Gloves

Instructions:

Before the lesson;

Pre-grate some milk and some white chocolate (keep them separate!) .

Cut up the crunchie and lion bars into small blocks so you can see inside them.

Set up a Bunsen, gauze, tripod and safety mat.

During the lesson:

Rock types:

Start with the bars. Get pupils to think about what they represent:

Lion bar = Sedimentary

Layered rock which sometimes contains fossils.

Marble effect chocolate = Metamorphic

Mixture of the other rock types that have been put though intense heat and pressure deep under the ground

Crunchie bar = Igneous

Rock that has cooled from molten lava that contains crystals and often air bubbles.

Remind higher pupils that quick cooling lava = small crystals and slow cooling lava = large crystals

Choc Cycle:

Grate some chocolate and ask pupils what it's supposed to represent (weathering) .

Sprinkle a layer of milk chocolate followed by a layer of white chocolate and ask pupils what it is supposed to represent (erosion and transport).

Repeat layering of milk then white chocolate.

Ask pupils what needs to happen in order to turn the sediment into rock (Burial and pressure)

Apply pressure with gloved hands or utensil.

Ask pupils what rock type this is (sedimentary) and what features does it have? (layers and sometimes fossils)

Ask what rock type comes next (metamorphic) and what needs to happen to make metamorphic? (extreme heat and pressure underground)

Apply heat by putting the bowl on top of the tripod with gentle heat from the Bunsen. Here you can ask pupils to identify issues with this stage of the model (no pressure and not extreme heat). DO NOT let it melt! Otherwise you have made igneous when it cools!

Show pupils the marble effect and ask about the features of metamorphic rock and how it comes to be that way (it is subject to heat and pressure so the layers mix and look 'marbled' and it cools because it is pushed towards the earth's surface due to uplift)

Remind pupils that this rock is then weathered erodes, and then turned into sediment rock again.

It may then be melted and turned into lava –so put the bowl back over the Bunsen to melt the chocolate. Ask what happens next (it cools – becoming rock)

What rock type is this? (Igneous) what features does it have? (crystals and air bubbles).

Remind pupils that this rock is then weathered erodes, and then turned into sediment rock again.

Perks:

If you used the gloves and clean utensils you can share the chocolate out with the class!

Safety:

Use a screen or pupils wear goggles when heating the chocolate.

Check ingredients in advance for ‘contains nuts or traces of nuts’.

Glass bowl gets very hot!